

PHTRP Cohort 004 – Prophage Analysis Practical Assignment

Title: Identifying Prophage Regions Using NCBI and ProkSee

Scenario

You have joined a microbiology laboratory investigating bacterial genomes. Your supervisor has asked you to check whether one bacterial genome contains integrated bacteriophages (prophages). Use NCBI to retrieve the genome and ProkSee (PHASTEST) to identify potential prophage regions.

Instructions

- Search for Escherichia coli K-12 MG1655 in the NCBI Genome database.
- Filter for Complete genomes and open the reference genome.
- Record the chromosome accession number (NC_000913.3).
- Open ProkSee, create a new map, and run the PHASTEST tool within it
- Answer the questions below.

Questions

1. How many prophage regions were identified? **4 regions**
2. How many of the prophage regions were classified as Intact? **2 regions (region 2 and region 4)**
3. Select one intact prophage and record the following:
 - Region number - **region 4**
 - Coordinates - **start coordinate 1,618,855 bp - end coordinate 1,653,915 bp**
 - Length - **35061 bp**
4. From the Details tab, identify three phage-associated genes found within your selected prophage.

For each gene, report:

	Gene name	Function	E-value
phage	PP_01553	Putative exonuclease	0.0000373
phage	PP_01554	IS2 transposase InsD	1.64e-133
phage	PP_01555	Putative integrase	1.32e-173

5. In your own words, explain why an integrase is considered strong evidence that a region is a prophage.

The integrase enzyme provides strong proof that a genomic area is prophage-related because it is an enzyme that specifically integrates phage DNA into the host chromosomes. This enzyme can identify specific attachment sites and catalyse the insertion of phage DNA into the bacterium's genome, allowing a bacteriophage to enter a lysogenic state. Therefore, finding an integrase enzyme within a predicted phage-like region strongly supports that the region is from a phage.

6. Explain the meaning of the E-value for one of the genes you reported.

The E-value is a statistical measure used to indicate how likely it is that a sequence match found during genome annotation occurred by chance. For instance, the E-value of 1.64e-133 for the phage provides strong evidence that the sequence is homologous to a known phage protein and that the match is unlikely to have occurred by chance.